Orthogonalization

Recall from last class:

- A set of vectors $\{u_1, \ldots, u_p\}$ in $\mathbb{R}^n$ is **orthogonal** if $u_k \cdot u_j = 0$ for all $j \neq k$.
- **Theorem 6.3** Let A be an $m \times n$ matrix, then

$$(\text{Row}(A))^\perp = \text{Null}(A) \text{ and } (\text{Col}(A))^\perp = \text{Null}(A^T).$$

- **Theorem 6.4:** If $S = \{u_1, \ldots, u_p\}$ is an orthogonal set of non-zero vectors in $\mathbb{R}^n$, then S is linearly independent and hence a basis for Span S.
- An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.
- **Theorem 6.5:** Let $\{u_1, \ldots, u_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $y \in W$, then

$$y = c_1 u_1 + \cdots + c_p u_p$$

  where

$$c_j = \frac{y \cdot u_j}{u_j \cdot u_j}.$$

- Let u be a non-zero vector in $\mathbb{R}^n$ and let v be a vector in $\mathbb{R}^n$, then the **orthogonal projection** of v onto u, denoted by $\text{proj}_u v$, is given by

$$\text{proj}_u v = \frac{v \cdot u}{u \cdot u} u$$

  Let $L = \text{Span}\{u\}$, then we also write $\text{proj}_L v$ for the orthogonal projection of v onto the line defined by the span of u; namely

$$\text{proj}_L v = \text{proj}_u v.$$

- **Note:** Theorem 6.5 says that

$$y = \text{proj}_{u_1} y + \text{proj}_{u_2} y + \cdots + \text{proj}_{u_p} y.$$

- A set $\{u_1, \ldots, u_p\}$ is **orthonormal** if it is orthogonal and $\|u_k\| = 1$ for all k.
- **Theorem 6.6:** An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I_n$.
- **Theorem 6.7:** Let U be an $m \times n$ matrix with orthonormal columns. Let $x, y \in \mathbb{R}^n$. Then
  1. $\|Ux\| = \|x\|$.
  2. $(Ux) \cdot (Uy) = x \cdot y$.
  3. $(Ux) \cdot (Uy) = 0$ if and only if $x \cdot y = 0$.
- An **orthogonal matrix** U is a square matrix where $U^{-1} = U^T$.

Orthogonal Decomposition Theorem: Let W be a subspace of $\mathbb{R}^n$. Then each $y \in \mathbb{R}^n$ can be uniquely written as

$$y = \hat{y} + z$$

where $\hat{y} \in W$ and $z \in W^\perp$.

Moreover, if $\{u_1, \ldots, u_p\}$ is any orthogonal basis for W, then

$$\hat{y} = \left(\frac{y \cdot u_1}{u_1 \cdot u_1}\right) u_1 + \cdots + \left(\frac{y \cdot u_p}{u_p \cdot u_p}\right) u_p,$$

and $z = y - \hat{y}$.

Definition 1. *The vector $\hat{y}$ is the* **orthogonal projection** *of y onto the subspace W. Written as*

$$\hat{y} = proj_W y.$$

1. Prove the Theorem.

Best Approximation Theorem: Let W be a subspace of $\mathbb{R}^n$. Let $\hat{y}$ be the orthogonal projection of y onto W, then

$$\|y - \hat{y}\| < \|y - v\|$$

for all $v \in W \setminus \{\hat{y}\}$.

2. Prove the Theorem.

Theorem 6.10: If $\{u_1, \ldots, u_p\}$ is an orthonormal basis for W, then

$$\text{proj}_W y = (y \cdot u_1)u_1 + \cdots + (y \cdot u_p)u_p.$$

If $U = [u_1 \ \ldots \ u_p]$, then

$$\text{proj}_W y = UU^T y.$$

3. Prove the theorem.

Question: How can we find an orthogonal basis for a given subspace?

4. Let

$$W = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Find an orthogonal basis for W.

Gram-Schmidt Orthogonalization. Given a basis $\{x_1, \ldots, x_p\}$ for a non-zero subspace W of $\mathbb{R}^n$, define

$$\begin{aligned}
v_1 &= x_1 \\
v_2 &= x_2 - \left(\frac{x_2 \cdot v_1}{v_1 \cdot v_1}\right) v_1 \\
v_3 &= x_3 - \left(\frac{x_3 \cdot v_1}{v_1 \cdot v_1}\right) v_1 - \left(\frac{x_3 \cdot v_2}{v_2 \cdot v_2}\right) v_2 \\
&\vdots \\
v_p &= x_p - \left(\frac{x_p \cdot v_1}{v_1 \cdot v_1}\right) v_1 - \cdots - \left(\frac{x_p \cdot v_{p-1}}{v_{p-1} \cdot v_{p-1}}\right) v_{p-1}.
\end{aligned}$$

Then $\{v_1, \ldots, v_p\}$ is an orthogonal basis for W. In addition $\text{Span}\{v_1, \ldots, v_k\} = \text{Span}\{x_1, \ldots, x_k\}$ for all $1 \leq k \leq p$.

5. Prove the Theorem.

QR Factorization. If A is an $m \times n$ matrix with linearly independent columns, then $A = QR$ where Q is an $m \times n$ matrix whose columns form an orthonormal basis for the column space of A and R is an upper triangular $n \times n$ matrix with positive entries along the diagonal.

6. Prove the Theorem.

7. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. Find QR factorization of A.

Orthogonalization – Answers / Solutions

1. *Proof.* Suppose $\{u_1, \ldots, u_p\}$ is an orthogonal basis for W, then define

$$\hat{y} = \left(\frac{y \cdot u_1}{u_1 \cdot u_1}\right) u_1 + \cdots + \left(\frac{y \cdot u_p}{u_p \cdot u_p}\right) u_p,$$

and $z = y - \hat{y}$. We first prove that z is orthogonal to W. Then

$$z \cdot u_k = (y - \hat{y}) \cdot u_k = y \cdot u_k - \hat{y} \cdot u_k = y \cdot u_k - \frac{y \cdot u_k}{u_k \cdot u_k} u_k \cdot u_k = .$$

Thus z is orthogonal to a basis for W, and hence z is orthogonal to all of W.

We need to prove uniqueness. Suppose $y = \hat{y}_1 + z_1$ for $\hat{y}_1 \in W$ and $z_1 \in W^\perp$. Thus we have $\hat{y} + z = \hat{y}_1 + z_1$, and hence $\hat{y} - \hat{y}_1 = z_1 - z$. Sinice W is a subspace, $\hat{y} - \hat{y}_1$ is in W. Since $\hat{y} - \hat{y}_1 = z_1 - z$, $\hat{y} - \hat{y}_1$ is in $W^\perp$. Thus $(\hat{y} - \hat{y}_1) \cdot (\hat{y} - \hat{y}_1) = 0$, and hence $(\hat{y} - \hat{y}_1) = 0$ and $\hat{y} = \hat{y}_1$. Solving for z, we get $z = z_1$. Thus $y = \hat{y} + z$ is the unique way to write y as the sum of an element of W and an element of $W^\perp$. □

2. *Proof.* Let $v \in W \setminus \{\hat{y}\}$, then

$$\|y - v\|^2 = \|y - \hat{y} + \hat{y} - v\|^2 .$$

Since $\hat{y} - v \in W$ and $y - \hat{y} \in W^T$, we get $\hat{y} - v$ and $y - \hat{y}$ are orthogonal. Thus by the Pythagorean theorem, we get

$$\|y - v\|^2 = \|y - \hat{y} + \hat{y} - v\|^2 = \|y - \hat{y}\|^2 + \|\hat{y} - v\|^2 .$$

Since $v \in W \setminus \{\hat{y}\}$, we know $\|\hat{y} - v\| > 0$. Thus

$$\|y - v\|^2 > \|y - \hat{y}\|^2 ,$$

and taking square roots we get

$$\|y - v\| > \|y - \hat{y}\| .$$

□

3. *Proof.* Since the vectors u_k all have length one, the usual orthogonal projection formula gives

$$\text{proj}_W y = (y \cdot u_1) u_1 + \cdots + (y \cdot u_p) u_p.$$

If $U = [u_1 \ \ldots \ u_n]$, then

$$UU^T y = \begin{bmatrix} u_1 & \ldots & u_p \end{bmatrix} \begin{bmatrix} u_1^T \\ \vdots \\ u_p^T \end{bmatrix} y = \begin{bmatrix} u_1 & \ldots & u_p \end{bmatrix} \begin{bmatrix} u_1 \cdot y \\ \vdots \\ u_p \cdot y \end{bmatrix} = (u_1 \cdot y) u_1 + \cdots + (u_p \cdot y) u_p = \text{proj}_W y.$$

□

4. This example is from the textbook. The solution presented here is essentially the same. Let x_1, x_2, x_3 be the three basis vectors for W.

Let $v_1 = x_1 = \begin{bmatrix}1\\1\\1\\1\end{bmatrix}$. Now we want a vector $v_2 \in W$ that is orthogonal to v_1. Thus we subtract the orthogonal projection of the second basis vector onto v_1 to get the orthogonal component of the second basis vector; namely,

$$v_2 = x_2 - \text{proj}_{v_1} x_2 = \begin{bmatrix}0\\1\\1\\1\end{bmatrix} - \frac{3}{4}\begin{bmatrix}1\\1\\1\\1\end{bmatrix} = \begin{bmatrix}-\frac{3}{4}\\\frac{1}{4}\\\frac{1}{4}\\\frac{1}{4}\end{bmatrix}.$$

It seems the subsequent computations will be more pleasant if we scale v_2 to have integer entries, so we take $v_2' = 3v_2$.

Now we want a third vector $v_3 \in W$ such that v_2 is orthogonal to v_1 and v_2. Again we substract the amount of x_3 in the direction of v_1 and v_2; namely,

$$v_3 = x_3 - \text{proj}_{\text{Span}\{v_1,v_2\}} x_3 = x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2'}{v_2' \cdot v_2'} v_2' = x_3 - \frac{1}{2} v_1 - \frac{2}{12} v_2' = \begin{bmatrix}0\\-\frac{2}{3}\\\frac{1}{3}\\\frac{1}{3}\end{bmatrix}.$$

Now v_1, v_2, v_3 form an orthogonal basis for W.

5. The following is the proof from Lay.

Proof. For $1 \le k \le p$, let $W_k = \text{Span}\{x_1, \dots, x_k\}$. Set $v_1 = x_1$. Then $W_1 = \text{Span}\{v_1\}$.

For $k < p$, $\{v_1, \dots, v_k$ is an orthogonal basis for W_k. Define

$$v_{k+1} = x_{k+1} - \text{proj}_{W_k} x_{k+1}.$$

By the Orthogonal Decomposition Theorem, v_{k+1} is orthogonal to W_k and $v_{k+1} \in W_{k+1}$. Note $v_{k+1} \neq 0$, so $\{v_1, \dots, v_{k+1}\}$ is an orthogonal basis for W_{k+1}. We continue the process until $k + 1 = p$. $\square$

6. *Proof.* The columns of A form a basis for $\text{Col}(A)$. Apply Gram-Schmidt to get an orthonormal basis $u_1, \dots, u_n$ for the column space of A. Let $Q = [u_1 \ \dots \ u_n]$. For each k,

$$x_k \in \text{Span}\{u_1, \dots, u_k\},$$

and hence there exists scalars $r_{1k}, \dots r_{kk}$ so that

$$x_k = r_{1k} x_1 + \cdots + r_{kk} x_k.$$

Now $x_k = Qr_k$ where $r_k = \begin{bmatrix} r_{1k} \\ \vdots \\ r_{kk} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$. Define $R = [r_1 \ \ldots \ r_n]$, then

$$A = [x_1 \ \ldots \ x_n] = [Qr_1 \ \ldots \ Qr_n] = QR.$$

$\square$

7. From Gram-Schmidt, we have an orthogonal basis

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix} \right\},$$

and scaling to get an orthonormal basis, we get

$$\left\{ \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \begin{bmatrix} -3/\sqrt{12} \\ 1/\sqrt{12} \\ 1/\sqrt{12} \\ 1/\sqrt{12} \end{bmatrix}, \begin{bmatrix} 0 \\ -2\sqrt{6} \\ 1/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix} \right\}.$$

Thus $Q = \begin{bmatrix} 1/2 & -3/\sqrt{12} & 0 \\ 1/2 & 1/\sqrt{12} & -2\sqrt{6} \\ 1/2 & 1/\sqrt{12} & 1/\sqrt{6} \\ 1/2 & 1/\sqrt{12} & 1/\sqrt{6} \end{bmatrix}$. Since $A = QR$ and Q has orthonormal columns, we get $R = Q^T A$. Thus $R = \begin{bmatrix} 2 & 3/2 & 1 \\ 0 & 3/\sqrt{12} & 2/\sqrt{12} \\ 0 & 0 & 2/\sqrt{6} \end{bmatrix}$.